

Procurement Bill of Materials

Module-level ordering guide (no vendor links)

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Platform: esp32

Total line items: 13

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Procurement note: Verify connector orientation, voltage levels, and form factor before ordering.

#	Item	Qty	Exactness	Ref	Required specs
1	ESP32 DevKit v1	1	Compatible	esp32	This project targets the ESP32 DevKit v1 (Espressif · esp32). • Match USB connector style and pinout.
2	MCP2515 CAN Bus Module (TJA1050)	1	Compatible	MCP2515	Interface: SPI • Power: 5V • Module form factor varies by seller; match pin labels and connector layout.
3	BNO055 9-Axis IMU	1	Compatible	BNO055	Interface: I2C • Power: 3.3V
4	DRV8825 Stepper Driver	1	Compatible	DRV8825	Module form factor varies by seller; match pin labels and connector layout.
5	Nema 17	1	Compatible	NEMA17	NEMA 17-size bipolar stepper motor for motion axes. Typical hobby modules are 42mm-frame, 4-wire/two-phase motors with 1.8 degree step angle (200 steps/rev); use with a current-limited stepper driver such as TMC2209/A4988/DRV8825 rather than direct GPIO.
6	BTS7960 43A H-Bridge Motor Driver	1	Compatible	BTS7960	Power: 5V • Module form factor varies by seller; match pin labels and connector layout.
7	BTS7960 43A H-Bridge Motor Driver	1	Compatible	BTS7960	Power: 5V • Module form factor varies by seller; match pin labels and connector layout.
8	High-Torque DC Motor (Left Track)	1	Generic	—	High-torque DC motor driving the left caterpillar track. Powered via BTS7960 H-bridge driver.
9	High-Torque DC Motor (Right Track)	1	Generic	—	High-torque DC motor driving the right caterpillar track. Powered via BTS7960 H-bridge driver.
10	MP1584 Buck Converter	1	Compatible	MP1584	Power: 5V
11	4S LiPo Battery 14.8V 5000mAh	1	Generic	—	Power: 14.8V
12	RPLiDAR A2M1 2D Laser Scanner	1	Compatible	A2M1	Slamtec RPLiDAR A2M1 360° 2D laser distance scanner. Communicates via USB to Jetson Nano USB Port 1 for high-bandwidth polar scan data. 5V USB-powered. Connects to Jetson Nano — outside the ESP32 microcontroller project boundary.
13	Binocular Stereo Depth Camera (USB 3.0)	1	Generic	—	USB 3.0 stereo depth camera for real-time depth mapping. Connects to Jetson Nano USB 3.0 Port 2 for high-bandwidth stereo frame streaming. Outside the ESP32 microcontroller project boundary.